

From Parcels to Blocks: Rescaling Deep Reinforcement Learning for Farmland Consolidation Planning through Spatial Abstraction

[Anonymous Author 1]^a, [Anonymous Author 2]^{a,*}

^a[*Anonymous Institution*], [*Anonymized*]

Abstract

Real cadastral land-consolidation problems require decision support that handles irregular parcels, sparse adjacencies, heterogeneous areas, and policy thresholds for contiguous operating fields. We propose a block-level spatial abstraction for farmland consolidation planning. Swappable farmland and forest parcels are aggregated into spatially coherent consolidation blocks using barrier-aware segmentation and constrained clustering. A Maskable PPO scheduler allocates a limited consolidation budget across blocks, while a connectivity-aware greedy engine executes parcel-level farmland–forest exchanges within selected blocks. The framework follows a macroscopic planning, microscopic execution design and treats *baimu fang* formation (100-mu consolidated fields) as a threshold-based consolidated-field metric evaluated with slope and contiguity. Experiments on three townships in Bishan District, Chongqing, using 3,607–12,950 Third National Land Survey parcels and 78–338 constructed blocks, show stable convergence across 15 seed-township combinations. The clearest scheduler-specific evidence occurs on the medium township: under the same within-block engine and reward, DRL is the only tested block-selection rule, including Reward-Greedy and finite-depth lookahead checks, to achieve positive *baimu fang* area growth (+12.9 ha versus negative baselines). On the small and large townships, reward-aware greedy planning remains competitive or stronger on several metrics. The contribution is therefore a scalable planning layer and evidence that learned scheduling can add value under specific multi-objective tension, rather than a gen-

*Corresponding author

Email address: [anonymized for peer review] ([Anonymous Author 2])

eral claim that reinforcement learning dominates heuristics. Because the experiments use one district and parcel-count-conserving rather than area-balanced swaps, the framework is best interpreted as a pre-screening and scenario-exploration tool.

Keywords: deep reinforcement learning, farmland consolidation, spatial abstraction, cadastral parcels, geospatial decision support, spatial optimization, baimu fang

1. Introduction

Farmland consolidation reorganizes fragmented agricultural parcels into contiguous, productive blocks. In China, high-standard farmland policy defines operational field-block targets through standards such as GB/T 30600-2022 [8] and national construction plans [14]. One common planning unit is the *baimu fang*, a 100-mu consolidated field (≥ 6.67 ha of contiguous farmland). In international terms, this is a threshold-based consolidated operating-field metric. In the mountainous regions of southwest China, where steep slopes, fragmented parcels, and interspersed forest are common, deciding which farmland and forest parcels to exchange is a large combinatorial problem.

Deep reinforcement learning (DRL) has been useful for sequential spatial planning, but its value for cadastral farmland consolidation is not obvious. Real parcels differ from grid cells in area, shape, adjacency, and barrier structure. These properties can make individual farmland–forest swaps nearly independent: large parcels dominate slope, sparse Queen contiguity limits neighborhood effects, and roads or waterways split the swappable graph into small components. A parcel-level MDP can therefore collapse toward a one-step ranking problem. We treat this as a design diagnosis, not a proof that all parcel-level DRL must fail.

We propose a **block-level spatial abstraction** that asks the agent to allocate a fixed consolidation budget across spatial units rather than individual parcels. Selecting one block consumes budget that cannot be used elsewhere, and swaps inside a block can affect adjacent blocks through township-wide contiguity and *baimu fang* metrics. The framework has three stages:

1. **Spatial restructuring:** Thousands of irregular parcels are aggregated into tens to hundreds of spatially coherent *consolidation blocks* using a

hybrid method combining natural barrier segmentation (roads, waterways, construction land) with constrained agglomerative clustering.

2. **Block-level MDP:** The optimization is formulated as a Markov Decision Process where the agent selects which block to invest consolidation resources in at each step, observing per-block characteristics and global state metrics.
3. **Connectivity-aware execution:** Within each selected block, a deterministic greedy engine executes farmland–forest swaps using a connectivity-aware scoring function that preferentially removes steep, isolated farmland and converts gentle, farmland-adjacent forest—creating spatial spillover effects across block boundaries.

This “*macroscopic planning, microscopic execution*” design mirrors consolidation governance: planners allocate quotas across project areas, while local implementers execute parcel-level conversions. Fig. 1 illustrates the architecture.

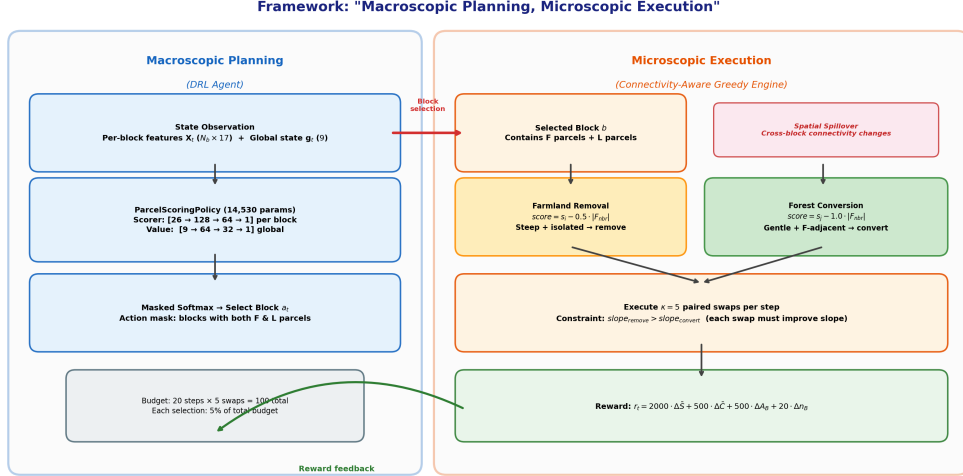


Figure 1: Block-level planning framework. The DRL scheduler allocates budget across blocks, and a deterministic greedy engine executes parcel-level swaps within selected blocks.

We evaluate the framework on three townships with 3,607–12,950 parcels and 78–338 constructed blocks. Five block-selection baselines share the same within-block engine, including a one-step Reward-Greedy baseline that uses

the same reward as DRL. We also run exact depth-two and pruned depth-three lookahead checks on Township B, where the key scheduler gap appears. The principal findings are:

1. The block-level MDP converges in all 15 seed-township combinations under identical hyperparameters.
2. The trained policies show township-specific profiles: balanced on Township A, *baimu fang*-area preserving on Township B, and count-focused on Township C.
3. On Township B, DRL is the only tested rule to achieve positive *baimu fang* area (+12.9 ha versus −101.9, −108.8, and −103.8 ha for Reward-Greedy, exact depth-two lookahead, and pruned depth-three lookahead). This is the clearest evidence for a scheduler-specific advantage.
4. On Townships A and C, Reward-Greedy is competitive with or stronger than DRL on several metrics. The contribution is therefore block-level abstraction with a learnable scheduler, not a general claim that RL dominates heuristics.
5. Across seeds, the trained policies are stable, with reward coefficient of variation below 1% on Township C.

These results support spatial abstraction as a practical design choice for scaling sequential optimization in farmland consolidation, while leaving parcel-level DRL, longer-horizon search, and area-balanced training as robustness priorities.

2. Related Work

2.1. Farmland consolidation optimization

Farmland consolidation has been studied with multi-objective evolutionary algorithms [6, 3, 4], hybrid genetic or game-theoretic methods [11], ant-colony and cellular-automata couplings [10], and integer programming [1]. These approaches are useful for land-use allocation, but most treat optimization as a static assignment problem. They rarely model the sequential allocation of limited consolidation effort across candidate project areas.

Chinese high-standard farmland policy emphasizes operational blocks such as *baimu fang*. GB/T 30600-2022 specifies field size, slope, contiguity, access, and irrigation requirements [8], while national plans set quantitative construction targets [14]. Existing optimization studies have not fully linked parcel exchange decisions to explicit consolidated-field formation, although land consolidation can improve agricultural productivity [17].

2.2. DRL for spatial optimization

DRL has been applied to urban spatial planning [18], urban generation [16], and ecological connectivity planning [7]. These studies show that learned policies can support spatial configuration tasks, but they do not address rural cadastral consolidation where irregular parcels, hard land-type constraints, and policy-defined field-block targets interact. The gap addressed here is the combination of block-level sequential allocation, deterministic parcel-level execution, and *baimu fang* metrics on real cadastral data.

2.3. Hierarchical and multi-scale optimization

Spatial optimization can be decomposed across scales. Hierarchical reinforcement learning decomposes tasks into sub-goals [2], and spatial planning studies have coupled optimization scales through cellular automata [10]. Our design uses a simpler split: the agent chooses *where* to invest, and a deterministic engine decides *how* to execute parcel swaps.

2.4. Multi-objective and constrained reinforcement learning

The reward combines slope, contiguity, *baimu fang* area, and *baimu fang* count as a weighted scalar. This gives one policy-relevant operating point rather than a complete Pareto-conditioned policy. More formal multi-objective or constrained MDP formulations remain possible; here, the scalar reward keeps the planner interpretable and allows direct comparison with reward-aware greedy baselines.

2.5. Spatial aggregation and constrained regionalization

Block construction is related to constrained regionalization and zoning in GIScience. Methods such as SKATER, REDCAP, and constrained agglomerative clustering produce contiguous regions under adjacency constraints. We use a natural-barrier graph, with roads, waterways, and construction land excluded, so the resulting blocks have a physical planning interpretation rather than only a statistical one.

3. Study Area and Data

3.1. Study area

The study area is Bishan District (29.4°–29.7°N, 106.0°–106.3°E), Chongqing Municipality, southwest China. It lies between the Sichuan Basin

and surrounding low mountains, with elevations of 180–885 m. For double-anonymized review, the three experimental townships are reported as Township A, Township B, and Township C.

Three townships were selected to represent a gradient of spatial complexity:

- **Township A** (code 500227109): 3,607 parcels, 1,601 swappable (1,130 farmland + 471 forest), aggregated into 78 blocks.
- **Township B** (code 500227108): 5,500 parcels, 2,669 swappable (1,998 farmland + 671 forest), aggregated into 132 blocks.
- **Township C** (code 500227105): 12,950 parcels, 6,787 swappable (5,032 farmland + 1,755 forest), aggregated into 338 blocks.

Table 1: Township characteristics and block-level aggregation. “Init. *baimu*” is the initial number of qualifying 6.67-ha contiguous farmland components.

Township	Total	Swappable	Farmland	Forest	Blocks	Parcels/block	Median area (ha)	Init. <i>baimu</i>
A (Small)	3,607	1,601	1,130	471	78	20	33	9
B (Medium)	5,500	2,669	1,998	671	132	20	28	6
C (Large)	12,950	6,787	5,032	1,755	338	20	20	6

3.2. Data sources

Cadastral data. Third National Land Survey parcels for Bishan District were classified by DLBM code as farmland (011–013), forest (031–033), or other. Farmland and forest parcels are swappable; other parcels act as fixed barriers. The controlled TNLS data contain owner-sensitive geometries and are not redistributed. We release aggregate statistics, derived metrics, logs, model artifacts, and figure-generation scripts.

Geometry and terrain. Polygons were repaired, filtered for valid land-use codes, and re-projected to EPSG:32648. Areas were computed as planar polygon areas, with $A_{\text{baimu}} = 66,700 \text{ m}^2$. Slope was derived from Copernicus DEM GLO-30 [5] using zonal statistics, assigning each parcel a mean slope in degrees. Queen contiguity was computed on the swappable subset using `libpysal`; excluded roads, water, and construction land therefore segment the graph.

4. Methodology

4.1. Problem formulation

The farmland consolidation optimization is formulated as a tri-objective combinatorial problem. Given a set of parcels $\mathcal{P} = \{p_1, \dots, p_M\}$ with land-use types $l_i \in \{\text{farmland}, \text{forest}, \text{other}\}$, slope values s_i , and areas a_i , the goal is to find a land-use assignment $\mathbf{I}^*$ that simultaneously:

$$\text{Minimize: } \bar{S} = \frac{\sum_{i:l_i=F} s_i \cdot a_i}{\sum_{i:l_i=F} a_i} \quad (\text{area-weighted mean farmland slope}) \quad (1)$$

$$\text{Maximize: } \bar{C} = \frac{1}{|\mathcal{P}_F|} \sum_{i \in \mathcal{P}_F} \frac{|\{j \in \mathcal{N}(i) : l_j = F\}|}{|\mathcal{N}(i)|} \quad (\text{farmland contiguity}) \quad (2)$$

$$\text{Maximize: } B = \sum_{k=1}^K \mathbb{I}[\text{area}(\mathcal{C}_k) \geq A_{\text{baimu}}] \quad (\text{baimu fang count}) \quad (3)$$

where $\mathcal{N}(i)$ is the Queen-contiguity neighborhood, $\mathcal{C}_k$ are farmland connected components, and $A_{\text{baimu}} = 66,700 \text{ m}^2$. The optimization uses a 100 paired-swap budget and conserves farmland parcel count. Eq. 3 is discrete and globally computed, so one parcel conversion can push a near-threshold connected component across the policy boundary.

4.2. Block-level spatial abstraction

The key methodological step is aggregating parcels into *consolidation blocks*, which reduce the action space, create budget competition among spatial units, and align decisions with project-scale planning.

4.2.1. Hybrid barrier segmentation and clustering

Block construction has five steps. We first extract farmland and forest parcels as swappable candidates, then build a Queen-contiguity graph after excluding roads, waterways, construction land, and other fixed barriers. Breadth-first search identifies barrier-delimited connected components. Components with more than 30 parcels are subdivided by Ward-linkage agglomerative clustering with a spatial connectivity constraint, targeting approximately 20 parcels per block. Blocks with fewer than 3 parcels or less than 0.5 ha are removed. Table 2 and Fig. 2 summarize the result.

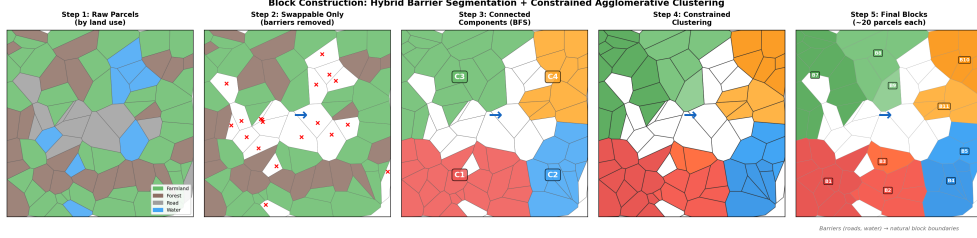


Figure 2: Block construction on synthetic parcels resembling Township A: swappable parcels are separated by barriers, decomposed into connected components, subdivided by constrained clustering, and labeled as final blocks.

Table 2: Block construction statistics. “Both F+F” is the share of blocks eligible for within-block swaps.

Township	Blocks	Assigned (%)	Both F+F (%)	Median parcels/block	Median area (ha)
A (78 blocks)	78	98	99	20	33
B (132 blocks)	132	99	91	20	28
C (338 blocks)	338	98	97	20	20

4.3. Block-level Markov Decision Process

The optimization is a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, \mathcal{T}, R, H)$ in which each action selects one block for investment.

4.3.1. State space

At step t , the state contains per-block features $\mathbf{X}_t \in \mathbb{R}^{N_b \times 17}$ and a global vector $\mathbf{g}_t \in \mathbb{R}^9$. The block features describe slope distribution, area, farmland/forest counts and ratios, within-block contiguity, compactness, neighboring-block context, remaining budget, and episode progress. The global vector records township-level farmland and forest slope, contiguity, farmland-count deviation, budget use, progress, current *baimu fang* count and area, and count change. The raw state dimensions are 1,335, 2,253, and 5,755 for Townships A–C; the shared-scorer policy keeps the number of learnable parameters independent of N_b .

4.3.2. Action space

At each step, the agent selects one block $a_t \in \{1, 2, \dots, N_b\}$. An action mask $\mathbf{m}_t \in \{0, 1\}^{N_b}$ ensures that only blocks containing both *remaining unused* farmland and forest parcels are available:

$$m_t(b) = \mathbb{1} \left[n_F^{(b, \text{unused})}(t) > 0 \wedge n_L^{(b, \text{unused})}(t) > 0 \right] \quad (4)$$

where $n_F^{(b, \text{unused})}$ and $n_L^{(b, \text{unused})}$ count unused farmland and forest parcels. A parcel-level **swapped** mask prevents repeated use of the same parcel, so the 100-swap budget corresponds to distinct paired exchanges.

4.3.3. Transition function: connectivity-aware greedy engine

Upon selecting block b , the environment executes up to $\kappa = 5$ farmland–forest swaps within that block.

Farmland removal scoring. For each farmland parcel i in block b :

$$\text{score}_{\text{remove}}(i) = s_i - \delta \cdot |\{j \in \mathcal{N}(i) : l_j = F\}| \quad (5)$$

where $\delta = 0.5$ penalizes removal of well-connected farmland.

Forest conversion scoring. For each forest parcel j in block b :

$$\text{score}_{\text{convert}}(j) = s_j - \gamma \cdot |\{k \in \mathcal{N}(j) : l_k = F\}| \quad (6)$$

where $\gamma = 1.0$ rewards forest parcels adjacent to farmland because the lowest score is converted. The 2:1 asymmetry between γ and δ encodes an active-consolidation bias: fill gaps near existing farmland while discouraging destruction of well-connected fields. These constants are fixed for all townships and baselines.

The farmland parcel with the highest $\text{score}_{\text{remove}}$ and the forest parcel with the lowest $\text{score}_{\text{convert}}$ are swapped, provided that the removed farmland’s slope exceeds the converted forest’s slope (otherwise the swap would worsen the mean farmland slope, and it is skipped). This process repeats κ times or until no improving swaps remain.

Spatial spillover. Contiguity and *baimu fang* metrics are computed on the full township graph, so a swap at a block boundary can change scores and connected components in adjacent blocks. This creates inter-step dependencies that a block-level scheduler can exploit.

4.3.4. Reward function

The reward combines four normalized improvement channels:

$$r_t = w_S \cdot \widetilde{\Delta S}_t + w_C \cdot \widetilde{\Delta C}_t + w_B \cdot \widetilde{\Delta A}_t^B + b_B \cdot \Delta n_t^{B,+} \quad (7)$$

where, denoting the area-weighted mean farmland slope by $\bar{S}_t$, the contiguity index by $\bar{C}_t$, the total *baimu fang* area by A_t^B , the *baimu fang* count by n_t^B ,

and the values at $t=0$ by $\bar{S}_0, \bar{C}_0, A_0^F$ (initial total farmland area):

$$\widetilde{\Delta S}_t = \frac{\bar{S}_{t-1} - \bar{S}_t}{|\bar{S}_0| + \varepsilon}, \quad \widetilde{\Delta C}_t = \frac{\bar{C}_t - \bar{C}_{t-1}}{|\bar{C}_0| + \varepsilon}, \quad (8)$$

$$\widetilde{\Delta A}_t^B = \frac{A_t^B - A_{t-1}^B}{A_0^F + \varepsilon}, \quad \Delta n_t^{B,+} = \max(0, n_t^B - n_{t-1}^B). \quad (9)$$

The weights are $w_S = 2000$, $w_C = 500$, $w_B = 500$, and $b_B = 20$ for all townships and seeds. A -1.0 penalty discourages selecting blocks that yield no completed swaps. The engine exactly conserves farmland parcel count, but not farmland area; area-balance implications are audited in Section 5.7 and discussed as a limitation.

4.3.5. Episode structure

Each episode has $H = 20$ steps and $\kappa = 5$ swaps per selected block, for a 100-swap budget. This setting balances within-block impact against enough macroscopic decisions for sequencing. A sensitivity check over $\kappa \in \{1, 3, 5, 10, 20\}$ is reported in Section 5.9. The discount factor is $\gamma_{\text{RL}} = 0.99$.

4.4. Baimu fang computation

Baimu fang identification is performed after each step by breadth-first search on the full township farmland graph. Each connected component whose area is at least $66,700 \text{ m}^2$ is counted, and both count and total qualifying area are returned. The largest township requires approximately 1.3 ms per call.

4.5. Policy network architecture

The *BlockScoringPolicy* evaluates each block with shared weights, so learnable parameters do not depend on N_b . It has a scorer and a value network.

Scorer network. For each block, the 17 local features are concatenated with the 9 global features and mapped to a scalar logit:

$$\text{logit}_b = \text{Scorer}([\mathbf{x}_b; \mathbf{g}_t]) \in \mathbb{R}, \quad \text{Scorer} : \mathbb{R}^{26} \rightarrow \mathbb{R} \quad (10)$$

implemented as a $26 \rightarrow 128 \rightarrow 64 \rightarrow 1$ MLP with tanh activations. The N_b logits are passed through a masked softmax:

$$\pi(a_t = b \mid \mathbf{s}_t) = \frac{\exp(\text{logit}_b) \mathbb{I}[m_t(b) = 1]}{\sum_{b'=1}^{N_b} \exp(\text{logit}_{b'}) \mathbb{I}[m_t(b') = 1]}. \quad (11)$$

Each township is trained independently, but the architecture itself is dimension-invariant.

Value network. The state value depends only on global features:

$$V(\mathbf{s}_t) = \text{Value}(\mathbf{g}_t) \in \mathbb{R}, \quad \text{Value} : \mathbb{R}^9 \rightarrow \mathbb{R} \quad (12)$$

implemented as a $9 \rightarrow 64 \rightarrow 32 \rightarrow 1$ MLP. The model has 14,530 parameters and per-state inference is under 1 ms on the largest township. We use orthogonal initialization [15], with gain 0.01 for the scorer and 1.0 for the value network.

4.6. Training

Models are trained with Maskable PPO from `sb3-contrib` [9]. Table 3 lists the hyperparameters.

Table 3: Training hyperparameters for the block-level MDP.

Hyperparameter	Value
Algorithm	Maskable PPO
Learning rate	1×10^{-3} (Adam)
Rollout length (n_{steps})	512
Minibatch size	256
PPO epochs	10
Discount factor (γ_{RL})	0.99
GAE parameter (λ)	0.95
Clip range	0.2
Entropy coefficient	0.01
Total timesteps	200,000
Episode length (H)	20 steps
Swaps per step (κ)	5
Total budget (B_{max})	100 paired swaps

For each township, 5 seeds are trained for 200,000 timesteps. This equals 10,000 episodes per seed and exceeds the observed $\sim 50,000$ -timestep plateau. Training on an NVIDIA A100 40GB took about 25 minutes (A), 50 minutes (B), and 2 hours (C) per seed.

4.7. Baseline methods

All baselines use the same 20-step, $\kappa = 5$, 100-swap block environment and the same within-block greedy engine. They differ only in block selection: Greedy-Global chooses the block with the largest one-step slope improvement; Greedy-Sequential follows blocks sorted by initial farmland slope; Random-Block samples valid blocks uniformly; Round-Robin cycles through blocks; Reward-Greedy simulates every unmasked block and selects the largest one-step multi-objective reward under Eq. 7. We also run Township B lookahead checks using the same reward and transition model: exact depth two and pruned depth three with beam width 8.

4.8. Evaluation

Trained models are evaluated with deterministic inference: at each step, the trained policy selects the block with the highest logit score (argmax over unmasked blocks). Each evaluation produces a single deterministic trajectory. We report slope change (%), contiguity change, *baimu fang* count change, *baimu fang* area change (ha), and budget utilization across all 5 seeds. Because the current swap engine conserves parcel count rather than total farmland area, these metrics are interpreted as scenario-screening indicators rather than legally area-balanced implementation outcomes; the area-balance implication is revisited in Section 6.4.

5. Results

5.1. Training convergence

All 15 seed-township combinations converge to positive reward plateaus, with visible stabilization by approximately 50,000 timesteps (Fig. 3; Table 4).

Table 4: Block-level training convergence under identical hyperparameters and 200,000 timesteps per seed.

Township	Block-level convergence
A (Small, 3,607 parcels, 78 blocks)	5/5 (100%)
B (Medium, 5,500 parcels, 132 blocks)	5/5 (100%)
C (Large, 12,950 parcels, 338 blocks)	5/5 (100%)
Total	15/15 (100%)

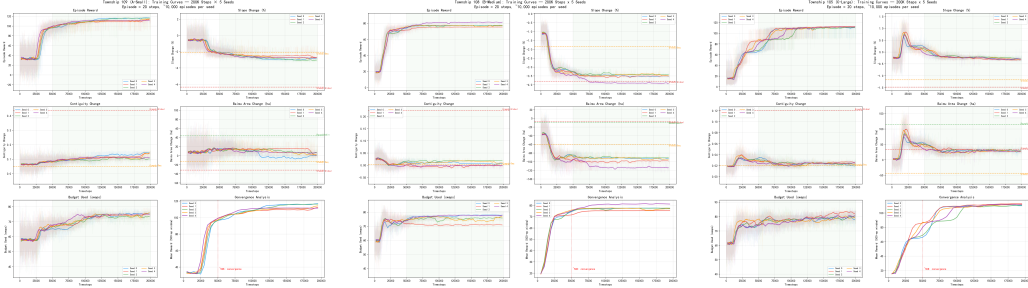


Figure 3: Training curves for Townships A–C, showing 5 seeds with a 200-episode moving average. Rewards stabilize by approximately 50K timesteps.

5.2. Cross-township comparison

Table 5 compares all primary methods.

Three patterns matter. First, DRL is the only tested method with positive *baimu fang* area on Township B ($+12.9 \pm 3.2$ ha); Reward-Greedy, exact depth-two lookahead, and pruned depth-three lookahead remain negative. This gap is therefore not explained by reward access alone or by the tested short-horizon search. Second, Reward-Greedy is competitive with or stronger than DRL on Townships A and C, so most of the broad multi-objective gain comes from block-level abstraction plus reward-aware selection. Third, Greedy-Global remains the strongest slope-only baseline but often damages *baimu fang* outcomes. Fig. 4 summarizes these trade-offs.

5.3. Township-specific strategy profiles

With identical hyperparameters, the trained policies express different township profiles. Township A is balanced but not Pareto-dominant. Township B trades slope reduction for contiguity and *baimu fang* area preservation; both DRL and Reward-Greedy make 20 block choices, but they choose different blocks. Reward-Greedy prioritizes high-slope blocks and dismantles existing consolidated fields, whereas DRL selects blocks that preserve or grow them (Fig. 5).

Township C shows a two-phase “invest-then-recover” trajectory: the agent first forms five new *baimu fang*, then recovers the slope channel to a final -0.33% (Fig. 8). Because Reward-Greedy forms six new fields on the same township, the strongest DRL-specific evidence remains Township B rather than Township C.

Table 5: Cross-township comparison. DRL reports mean $\pm$ std across 5 seeds. More negative slope is better; larger contiguity, *baimu fang* count, and *baimu fang* area are better.

Township	Method	Slope (%)	Cont.	Baimu cnt	Baimu ha
A (78 blk)	Greedy-Global	-6.36	+0.439	-2	-31.5
	Greedy-Seq	-2.12	+0.050	-1	-12.0
	Random (avg)	-0.79	+0.051	0	+1.4
	Round-Robin	-0.33	+0.087	-1	+43.8
	Reward-Greedy	-3.03	+0.198	0	+13.5
	DRL	-2.61 ± 0.44	$+0.130 \pm 0.025$	0	$+8.8 \pm 10.7$
B (132 blk)	Greedy-Global	-3.79	+0.226	-1	-6.9
	Greedy-Seq	-1.85	+0.003	0	-60.4
	Random (avg)	-1.31	+0.032	-0.6	-38.1
	Round-Robin	-0.28	+0.062	-1	-8.9
	Reward-Greedy	-4.24	+0.036	0	-101.9
	DRL	-1.64 ± 0.04	$+0.115 \pm 0.009$	-1	+12.9 $\pm$ 3.2
C (338 blk)	Greedy-Global	-1.48	+0.121	-1	+34.6
	Greedy-Seq	-1.17	+0.020	0	-43.0
	Random (avg)	-0.24	+0.026	+0.8	+9.6
	Round-Robin	+1.10	+0.047	+2	+114.4
	Reward-Greedy	-0.52	+0.045	+6	+28.0
	DRL	-0.33 ± 0.01	$+0.023 \pm 0.003$	$+5 \pm 0$	$+27.7 \pm 1.3$

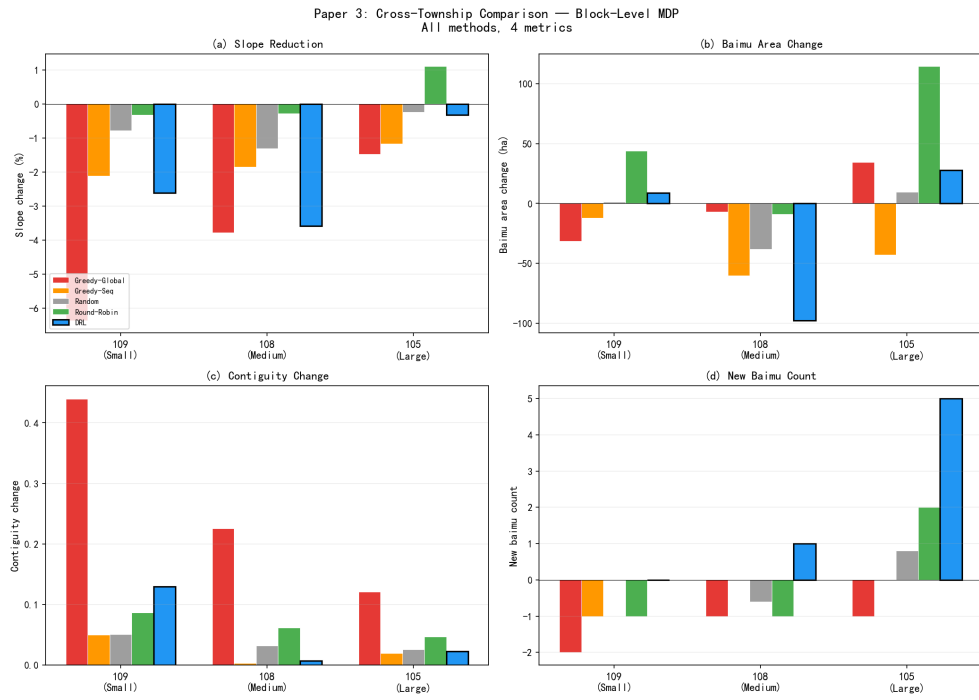


Figure 4: Cross-township comparison across slope, *baimu fang* area, contiguity, and *baimu fang* count. DRL is outlined in black.

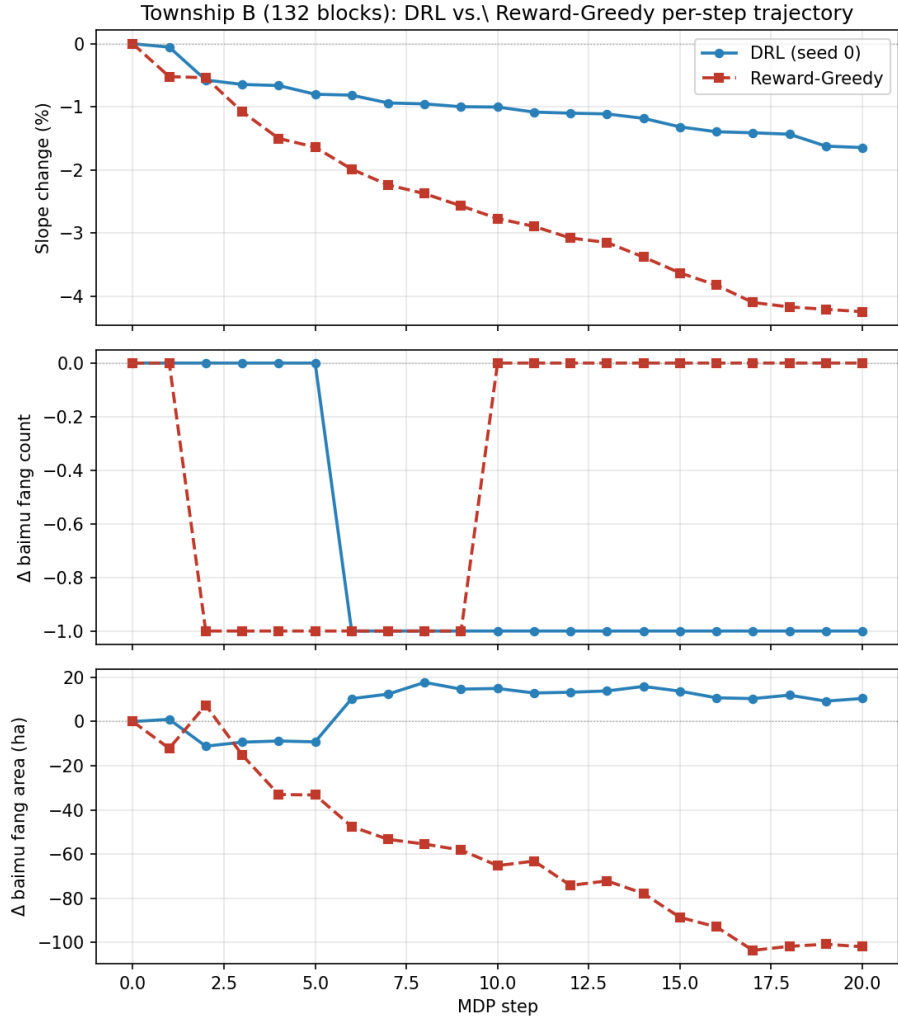


Figure 5: Township B trajectory comparison between DRL (seed 0) and Reward-Greedy. Reward-Greedy gains slope but loses *baimu fang* area, whereas DRL preserves and slightly grows it.

Table 6: Finite-depth lookahead checks on Township B under the same reward and transition model as Reward-Greedy.

Township	Method	Slope (%)	Cont.	Baimu cnt	Baimu ha
B	Lookahead d=2 b=0	-4.23	+0.013	+1	-108.8
B	Lookahead d=3 b=8	-4.26	+0.035	0	-103.8

5.4. Per-seed consistency

Table 7 reports per-seed DRL results.

Per-seed variability is low. Township C has reward CV of 0.26%, slope std of 0.01%, and *baimu fang* count identically +5 across all seeds. No township shows degenerate or bimodal seed behavior.

5.5. Multi-objective Pareto analysis

The Pareto plots reinforce the same interpretation. DRL and Reward-Greedy are the only primary methods that maintain non-negative *baimu fang* count on every township while also reducing slope. Reward-aware block selection defines most of the Pareto frontier; learned scheduling adds its clearest increment on Township B. Round-Robin maximizes *baimu fang* area on Township C, but only by increasing mean farmland slope.

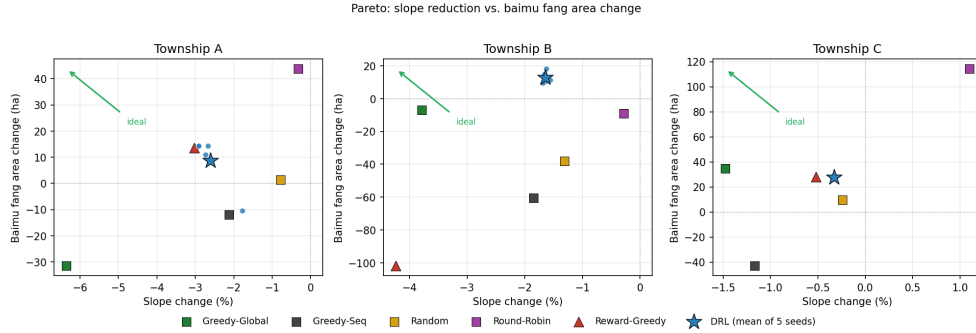


Figure 6: Pareto analysis of slope reduction and *baimu fang* area change. DRL is the only primary method with positive *baimu fang* area on Township B.

5.6. Post-hoc compactness audit of formed baimu fang

The main *baimu fang* count is topological: any Queen-connected farmland component above 6.67 ha qualifies. Because high-standard farmland

Table 7: Per-seed DRL results. CV is the coefficient of variation of total reward.

Township	Seed	Slope (%)	Cont.	Baimu cnt	Reward
A	0	-2.67	+0.149	0	154.1
	1	-1.78	+0.154	0	140.3
	2	-2.92	+0.092	0	161.2
	3	-2.92	+0.140	0	159.8
	4	-2.74	+0.113	0	152.7
	<i>CV</i>				<i>5.3%</i>
B	0	-1.64	+0.115	-1	37.6
	1	-1.56	+0.127	-1	38.7
	2	-1.63	+0.115	-1	42.0
	3	-1.66	+0.115	-1	40.5
	4	-1.69	+0.100	-1	35.6
	<i>CV</i>				<i>5.9%</i>
C	0	-0.32	+0.018	+5	265.8
	1	-0.35	+0.027	+5	267.0
	2	-0.33	+0.024	+5	266.6
	3	-0.34	+0.021	+5	265.2
	4	-0.33	+0.025	+5	266.3
	<i>CV</i>				<i>0.26%</i>

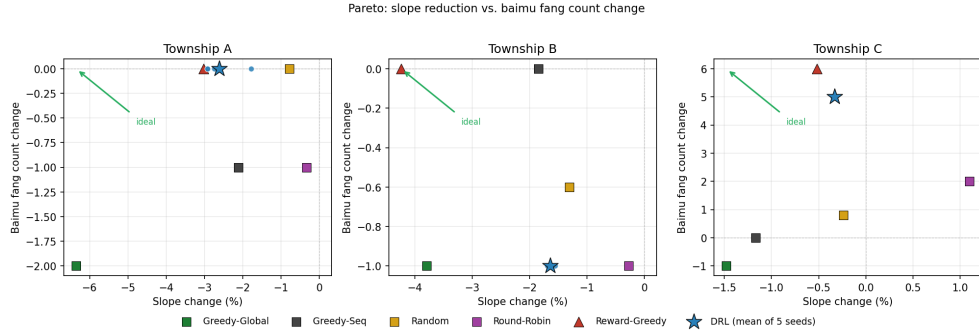


Figure 7: Pareto analysis of slope reduction and *baimu fang* count change. DRL and Reward-Greedy satisfy both non-negative count and negative slope across all townships.

also requires practical field shape, we audited final DRL components using isoperimetric quotient $IPQ = 4\pi A/P^2$ under three thresholds (Table 8).

Table 8: Post-hoc compactness audit of *baimu fang* formed by the block-level DRL agent. For each township and each evaluation seed we recover the final farmland configuration by replaying the agent’s block selection sequence, identify all connected farmland components with area ≥ 6.67 ha (the raw *baimu fang* count), and compute the isoperimetric quotient $IPQ = 4\pi A/P^2$ on the dissolved farmland polygon of each component ($IPQ=1$ for a circle; lower means more elongated/fragmented shape). The “Init. IPQ” columns report the same compactness counts on the *initial* land-use configuration before any optimization, providing a fairness baseline: the Queen-contiguity connectivity definition combined with real cadastral parcel shapes typically yields low-IPQ components even prior to consolidation, so the post-DRL counts at each τ should be read against the corresponding initial value rather than against an ideal of $\tau=1$. DRL values are means over 5 seeds.

Township	Initial state				After DRL optimization (mean over 5 seeds)				
	Init. raw	Init. IPQ ≥ 0.10	Init. IPQ ≥ 0.20	Init. IPQ ≥ 0.30	Final raw	Final IPQ ≥ 0.10	Final IPQ ≥ 0.20	Final IPQ ≥ 0.30	Δ raw
A	9	3	0	0	9.0	0.6	0.0	0.0	+0.0
B	6	1	0	0	5.0	1.6	0.0	0.0	-1.0
C	6	0	0	0	11.0	1.0	0.0	0.0	+5.0

The audit shows that raw topological counts are upper-bound estimates. Initial components are already irregular, no township reaches $IPQ \geq 0.20$, and DRL’s compactness gains are mixed because compactness is not part of the reward.

5.7. Farmland-area balance audit

The engine conserves farmland parcel count, not farmland area. We therefore audited final farmland-area drift for every method (Table 9).

DRL has small mean area drift on all townships (+0.13–+0.28%), so its reported gains are not driven by large farmland expansion. However, some baselines drift more, and Reward-Greedy on Township B loses both farmland area and *baimu fang* area. The present implementation is therefore a screening tool; deployment requires area-tolerant or exact area-balanced transitions.

5.8. Area-tolerance transition stress test

We also ran a Township B stress test with a transition engine that enforces cumulative farmland-area tolerance and accepts only slope-improving paired swaps. The DRL rows replay trained block histories under the modified transition; they are not retrained policies.

Table 9: Farmland-area balance audit. The optimization engine conserves farmland parcel count, not farmland area. This post-hoc audit replays each final configuration and reports the net farmland-area change. Random-Block and DRL are reported as mean $\pm$ standard deviation over five seeds; deterministic baselines report a single trajectory. Values are derived from controlled-access parcel geometry and released here only as aggregate diagnostics.

Township	Method	Δ farmland area (ha)	Δ farmland area (%)
A	Greedy-Global	-43.6	-2.06
A	Greedy-Sequential	-12.6	-0.60
A	Random-Block	-2.4 ± 13.0	-0.11 ± 0.61
A	Round-Robin	+31.2	+1.47
A	Reward-Greedy	+4.0	+0.19
A	DRL	$+2.9 \pm 9.6$	$+0.14 \pm 0.45$
B	Greedy-Global	-13.7	-0.40
B	Greedy-Sequential	-59.1	-1.74
B	Random-Block	-37.8 ± 25.9	-1.12 ± 0.77
B	Round-Robin	-11.6	-0.34
B	Reward-Greedy	-101.6	-3.00
B	DRL	$+9.6 \pm 4.0$	$+0.28 \pm 0.12$
C	Greedy-Global	+23.2	+0.40
C	Greedy-Sequential	-38.5	-0.67
C	Random-Block	$+7.7 \pm 30.2$	$+0.13 \pm 0.52$
C	Round-Robin	+89.8	+1.56
C	Reward-Greedy	+7.2	+0.12
C	DRL	$+7.4 \pm 1.9$	$+0.13 \pm 0.03$

Table 10: Township B area-tolerance transition check. The modified engine enforces cumulative farmland-area tolerance and uses an exact area-weighted mean-slope improvement check for paired swaps. DRL rows replay existing block histories only and are stress tests, not retrained policies.

Tolerance (%)	Method	Slope (%)	Cont.	Baimu cnt	Baimu ha	Farm area (%)	Budget
0.5	Greedy-Global	-3.79	+0.226	-1.0	-6.9	-0.40	100.0
0.5	Greedy-Sequential	-1.26	+0.027	+0.0	-14.1	-0.42	64.0
0.5	Round-Robin	-0.39	+0.045	-1.0	-14.1	-0.44	38.0
0.5	Reward-Greedy	-2.52	+0.123	+1.0	-13.3	-0.45	85.0
0.5	DRL replay mean	-1.93	+0.091	+0.0	-9.8	-0.36	77.6
1.0	Greedy-Global	-3.79	+0.226	-1.0	-6.9	-0.40	100.0
1.0	Greedy-Sequential	-1.78	+0.014	+0.0	-33.5	-0.92	65.0
1.0	Round-Robin	-0.46	+0.045	-1.0	-20.2	-0.61	45.0
1.0	Reward-Greedy	-3.12	+0.089	+1.0	-34.7	-0.99	80.0
1.0	DRL replay mean	-2.59	+0.081	+0.0	-29.7	-0.87	77.6

Under area tolerance, Reward-Greedy remains negative on *baimu fang* area, and replayed DRL histories also become negative. This does not invalidate the count-conserving experiment, but it shows that operational use must retrain and evaluate under the deployment transition rule.

5.9. Sensitivity to the swaps-per-step parameter κ

The swaps-per-step parameter κ controls the trade-off between many small block decisions and few large ones. We retrained Township A for $\kappa \in \{1, 3, 5, 10, 20\}$ with a fixed 100-swap budget and 50,000 timesteps per setting (Table 11).

The response is non-monotone. The extremes fail: $\kappa = 1$ commits only 8/100 swaps, while $\kappa = 20$ leaves too few macro-decisions. The middle range performs best, with $\kappa = 5$ giving the highest reward and $\kappa = 10$ giving the largest *baimu fang* area on Township A. The default is therefore balanced, not universally optimal.

6. Discussion

6.1. Why block-level abstraction can expose a useful scheduling problem

Block-level abstraction can expose scheduling value because it adds three structures that individual parcel swaps often lack. First, the 100-swap budget creates resource competition across blocks. Second, the connectivity-aware engine creates spatial spillover across block boundaries. Third, the 6.67 ha

Table 11: Sensitivity of the block-level DRL agent to the swaps-per-step parameter κ on Township A (78 blocks). The total paired-swap budget is fixed at 100; the MDP horizon is $H = 100/\kappa$. Each κ is trained for 50,000 timesteps with seed 0 on CPU and evaluated deterministically. “Budget used” is the number of paired swaps actually committed in the deterministic episode out of the 100-swap cap. Reward is the cumulative episodic return under the multi-component reward of Eq. 7 with the same weights used elsewhere in the paper. The $\kappa = 5$ value used in the main experiments achieves the highest cumulative reward (bold).

κ	H	Slope (%)	Cont.	Baimu cnt	Baimu ha	Budget used	Reward
1	100	−0.20	−0.011	+2	+6.7	8/100	−47.9
3	33	−2.07	+0.113	+1	−4.5	42/100	78.5
5	20	−2.01	+0.074	+1	+18.4	38/100	85.5
10	10	−0.59	+0.009	+1	+22.0	18/100	52.2
20	5	−0.06	+0.002	+1	−1.7	4/100	17.0

baimu fang threshold makes some block choices discontinuously valuable. These mechanisms explain why a learned scheduler can help in selected cases.

The claim remains bounded. The results show that this block-level formulation is trainable and interpretable, and that the Township B advantage is not reproduced by one-step Reward-Greedy, exact depth-two search, or pruned depth-three search. They do not prove that RL is uniquely necessary. Direct parcel-level DRL, value-guided rollouts, and longer-horizon search remain important comparisons.

6.2. The “invest-then-recover” strategy on Township C

Township C illustrates objective switching. In the first ten steps, the agent accepts an early slope penalty to push near-threshold clusters across the *baimu fang* threshold. In the remaining steps, it preserves the five new fields and recovers slope to a final −0.32% (Fig. 8).

This pattern is not unique evidence for DRL, because Reward-Greedy forms six new fields on Township C. It is useful mainly as an interpretable example of how block-level planning can shift between objectives within one episode.

6.3. Policy implications

The policy implication is decision support rather than automatic prescription. The block-level action resembles how planners allocate consolidation effort across project areas, and threshold-based metrics translate policy goals



Figure 8: Township C trajectory of the DRL agent. The policy first forms five new *baimu fang*, then recovers slope while holding the count fixed.

into computable objectives. However, cadastral parcels carry contracted management rights, and cross-ownership exchanges require negotiation and compensation [17]. The framework should therefore screen physical scenarios that planners can evaluate, not dictate implementable land adjustments. Administrative or cooperative boundaries could be added as hard barriers in block construction to reduce social transaction costs.

6.4. Limitations

Several limitations remain. First, parcel-count conservation is not area conservation. The DRL area-drift audit is small under the tested setting, but the area-tolerance stress test shows that deployment should train directly under an area-tolerant or exact area-balanced transition. Second, the study lacks a direct parcel-level DRL baseline and exhaustive long-horizon search; the reported finite-depth checks strengthen but do not close the planning-baseline space. Third, *baimu fang* identification is topological and does not yet enforce engineering shape requirements. Fourth, DRL does not dominate slope-only baselines, and the preferred reward weights depend on policy priorities. Fifth, block construction, (γ, δ) , and κ require broader sensitivity analysis. Finally, all townships are in one district, deterministic baselines lack uncertainty estimates, and DRL training is more costly than greedy planning. These boundaries support interpreting the framework as scalable scenario screening rather than a ready operational planner.

7. Conclusions

This paper proposed a block-level spatial abstraction for DRL-based farmland consolidation planning. Across three townships, the block-level MDP converged reliably and produced stable township-specific strategies. The strongest scheduler-specific result occurs on Township B, where DRL is the only tested method with positive *baimu fang* area growth while Reward-Greedy and finite-depth lookahead remain negative. On Townships A and C, however, Reward-Greedy is competitive or stronger on several metrics, showing that the main contribution is not a blanket claim that RL beats heuristics.

The defensible contribution is the combination of block-level abstraction, connectivity-aware within-block execution, and a learnable scheduler for cases with multi-objective tension. Because the current implementation

conserves parcel count rather than exact farmland area and uses a topological *baimu fang* definition, it should be read as a scenario-screening tool. Future work should prioritize area-balanced retraining, parcel-level and longer-horizon baselines, block-granularity sensitivity, and cross-district validation.

CRedit authorship contribution statement

[Anonymous Author 1]: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. [Anonymous Author 2]: Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The replication code for the block-level environment, training scripts, baselines, evaluation utilities, and figure-generation pipelines, as well as anonymized aggregate result files supporting all tables and figures in this paper, will be provided in an anonymized review repository at submission. The full Third National Land Survey (TNLS) cadastral dataset for Bishan District is governed by the access regulations of the relevant Chinese authorities and cannot be redistributed by the authors. Readers wishing to reproduce the experiments on the original data should obtain access through the proper channels of China’s Ministry of Natural Resources, after which the present codebase can ingest the data via the documented preprocessing pipeline. The repository includes derived block-level metrics, training logs, model artifacts, evaluation outputs, and figure-generation scripts sufficient to validate the reported tables and figures without redistributing the restricted raw parcels. Slope features were derived from the publicly available Copernicus DEM GLO-30 [5] via standard zonal statistics (mean slope per parcel, computed in the EPSG:32648 projection).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex and ChatGPT to support language editing, consistency checks, manuscript formatting, and submission-readiness auditing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] Aerts, J. C., Eisinger, E., Heuvelink, G. B., and Stewart, T. J. (2003). Using linear integer programming for multi-site land-use allocation. *Geographical Analysis*, 35(2):148–169.
- [2] Barto, A. G. and Mahadevan, S. (2003). Recent advances in hierarchical reinforcement learning. *Discrete Event Dynamic Systems*, 13(4):341–379.
- [3] Cao, K., Batty, M., Huang, B., Liu, Y., Yu, L., and Chen, J. (2011). Spatial multi-objective land use optimization: extensions to the non-dominated sorting genetic algorithm-II. *International Journal of Geographical Information Science*, 25(12):1949–1969.
- [4] Cao, K., Huang, B., Wang, S., and Lin, H. (2012). Sustainable land use optimization using boundary-based fast genetic algorithm. *Computers, Environment and Urban Systems*, 36(3):257–269.
- [5] European Space Agency (2021). Copernicus DEM GLO-30: Global 30 m Digital Elevation Model. European Space Agency, Copernicus Programme. Product handbook: Copernicus DEM Product Handbook, version 4.0. DOI: 10.5270/ESA-c5d3d65. Licence: ESA standard licence for Copernicus contributing missions.
- [6] Deb, K., Pratap, A., Agarwal, S., and Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197.
- [7] Equihua, J., Beckmann, M., and Seppelt, R. (2024). Connectivity conservation planning through deep reinforcement learning. *Methods in Ecology and Evolution*, 15(4):779–790.

- [8] Standardization Administration of China (2022). *GB/T 30600-2022: General Rules for High-Standard Farmland Construction* (Gaobiao zhun nongtian jianshe tongze). National Standard of the People’s Republic of China.
- [9] Huang, S. and Ontañón, S. (2022). A closer look at invalid action masking in policy gradient algorithms. In *The International FLAIRS Conference Proceedings*, volume 35.
- [10] Li, X., Yeh, A. G., and Liu, X. (2011). Coupling cellular automata and ant colony optimization for land-use allocation. *Environment and Planning B: Planning and Design*, 38(2):234–252.
- [11] Liu, Y., Tang, W., He, J., Liu, Y., Ai, T., and Liu, D. (2015). A land-use spatial optimization model based on genetic optimization and game theory. *Computers, Environment and Urban Systems*, 49:1–14.
- [12] Liu, Y., Fang, F., and Li, Y. (2014). Key issues of land use in China and implications for policy making. *Land Use Policy*, 40:6–12.
- [13] Long, H., Qu, Y., Tu, S., Zhang, Y., and Jiang, Y. (2018). Development of land use transitions research in China. *Journal of Geographical Sciences*, 28(10):1387–1399.
- [14] Ministry of Agriculture and Rural Affairs of the People’s Republic of China (2021). *National Plan for High-Standard Farmland Construction (2021–2030)*. Beijing.
- [15] Saxe, A. M., McClelland, J. L., and Ganguli, S. (2014). Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations*.
- [16] Wang, D., Fu, Y., Wang, P., Huang, B., and Lu, C. T. (2021). Automated urban planning for reimagining city configuration via adversarial learning: quantification, generation, and evaluation. *ACM Transactions on Spatial Algorithms and Systems (TSAS)*, 7(4):1–28.
- [17] Wu, Z., Liu, M., and Davis, J. (2005). Land consolidation and productivity in Chinese household crop production. *China Economic Review*, 16(1):28–49.

- [18] Zheng, Y., Lin, Y., Zhao, L., Wu, T., Jin, D., and Li, Y. (2023). Spatial planning of urban communities via deep reinforcement learning. *Nature Computational Science*, 3(9):748–762.